

Gender: Female | Birth year: 1975 | WHO: 1

Tumor: Lung adenocarcinoma | Lesions: Liver, Lung | Stage: IV

Clinical summary

Relevant systemic treatment history	2023-06 to 2025-01	Osimertinib
Relevant other oncological history	None	
Previous primary tumor	None	
Relevant non-oncological history	2023	Rheumatoid arthritis

Recent molecular results

Hartwig WGS (22-Feb-2025)

Molecular tissue of origin prediction	Lung: Non-small cell: LUAD (98%)
Tumor mutational load / burden	TML 160 / TMB 14 mut/Mb
Microsatellite (in)stability	Stable
HR status	Proficient (0)
Driver mutations	EGFR C797S, EGFR L858R, KRAS G12C, KRAS G12D
Amplified genes	None
Deleted genes	TP53 del
Homozygously disrupted genes	None
Gene fusions	MET(exon13)::MET(exon15) fusion
Driver virus	None
HLA-A	HLA-A*01:01, HLA-A*02:01

Trial-relevant IHC results

PD-L1	Score > 50%
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Standard-of-care options considered potentially eligible

There are no standard of care treatment options for this patient

Phase 2/3+ trials in NL that are open and potentially eligible (2 trials)

Trial	Cohort	Molecular	Sites	Warnings
METC 04 TEDR1 (Phase 2)	Lung cancer C797S cohort	EGFR C797S	NKI-AvL	None
EGFR-C797S-TRIAL (Phase 2)	EGFR C797S	EGFR C797S	Elisabeth-TweeSteden Ziekenhuis	

Trials matched solely on molecular event and tumor type (no clinical data used) are shown in italicized, smaller font.

Phase 1/2 (or unknown phase) trials in NL that are open and potentially eligible (3 trials)

Trial	Cohort	Molecular	Sites	Warnings
METC 02 KAYRAS (Phase 1/2)	Dose expansion - monotherapy - NSCLC	KRAS G12D, PD-L1 >= 50.0	Erasmus MC	Variant(s) G12D in KRAS but subclonal likelihood of > 50%

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Trial	Cohort	Molecular	Sites	Warnings
METC 01 IEMOEN (Phase 1)	Dose escalation - monotherapy (no slots)	None		Has not exhausted SOC (at least platinum doublet remaining)
EGFR-L858R-TRIAL (Phase 1)	<i>EGFR L858R</i>	<i>EGFR L858R</i>	<i>Elisabeth-TweeSteden Ziekenhuis</i>	

Trials matched solely on molecular event and tumor type (no clinical data used) are shown in italicized, smaller font.

International trials that are open and potentially eligible (2 trials)

Trial	Cohort	Molecular	Sites
EGFR-BE (Phase 1)	<i>EGFR L858R</i>	<i>EGFR L858R</i>	<i>Belgium: Brussels</i>
KRAS-G12C-TRIAL-DE (Phase 1)	<i>KRAS G12C</i>	<i>KRAS G12C</i>	<i>Germany: Stuttgart</i>

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Molecular Details

Hartwig WGS (EXAMPLE-LUNG-01-T, 22-Feb-2025)

General

Purity	Ploidy	TML Status	TMB Status	MS Stability	HR Status	DPYD	UGT1A1
50%	2.3	High (160)	High (14)	Stable	Proficient (0)	*1_HOM (Normal function)	*1_HOM (Normal function)

Predicted tumor origin

1. Lung: Non-small cell: LUAD	
Combined prediction score	98%
This score is calculated by combining information on:	
(1) SNV types	60%
(2) SNV genomic localisation distribution	70%
(3) Driver genes and passenger characteristics	80%

Other cohorts have a combined prediction of 2% or lower

Key drivers

Type	Driver	Trials (Locations)	Trials in Hartwig	Best evidence in External	Resistance in External
Mutation (gain of function)	EGFR L858R (2/4 copies)		NCT00000006, NCT00000007	Approved	
Mutation (gain of function)	EGFR C797S (1/4 copies)	TEDR1 (NKI-AvL)	NCT00000008	Pre-clinical	
Mutation (gain of function)	KRAS G12D (0.3/2 copies)*	KAYRAS (Erasmus MC)			
Mutation (gain of function)	KRAS G12C (0.3/2 copies)*		NCT00000009		

The table continues on the next page

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Gene and variant annotations and related content are powered by Genomenon Cancer Knowledgebase (CKB).

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Type	Driver	Trials (Locations)	Trials in Hartwig	Best evidence in External	Resistance in External
Deletion	TP53 del				
Known fusion	MET(exon13)::MET(exon15) fusion				
	Domain(s) kept: Tyrosine Kinase				
	Domain(s) lost: Juxtamembrane				

Other drivers or relevant events

Immunology

HLA gene	Type	Tumor copy number	Mutated in tumor
HLA-A	HLA-A*01:01	1.0	No
	HLA-A*02:01	0.0	No

PD-L1 **Score > 50%**

Event	Description	2025-02-22 Hartwig WGS
EGFR L858R (Tier I)	Mutation (gain of function)	VAF 0.5%
EGFR C797S (Tier II)	Mutation (gain of function)	VAF 0.25%

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Event	Description	2025-02-22 Hartwig WGS
KRAS G12C (Tier III)	Mutation (gain of function)	VAF 0.15%
KRAS G12D (Tier III)	Mutation (gain of function)	VAF 0.15%
MET(exon13)::MET(exon15) fusion (Tier III)	Known fusion Gain of function	Detected
TP53 del (Tier III)	Deletion Unknown protein effect	Detected
TMB		14.0
MSI		Stable

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Efficacy evidence

Standard of care options considered potentially eligible

The following standard of care treatment(s) could be an option for this patient. For further details per study see 'SOC literature details' section in extended report.

There are no standard of care treatment options for this patient

There are no standard of care treatment options for this patient

Resistance evidence

There are no standard of care treatment options for this patient

Treatment ranking

Event	Treatment	Score
EGFR L858R EGFR C797S	AFATINIB	2400
EGFR L858R	OSIMERTINIB	1900

On label clinical evidence

Event	CKB Event	Level A	Level B	Level C	Level D
EGFR C797S	EGFR C797S				AFATINIB <i>Lung non-small cell carcinoma (2015)</i>
EGFR L858R	EGFR L858R	OSIMERTINIB <i>Lung non-small cell carcinoma (2016)</i>	AFATINIB <i>Lung non-small cell carcinoma (2013)</i>		

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Off label clinical evidence

None

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Clinical Details

Clinical summary

Relevant systemic treatment history	2023-06 to 2025-01	Osimertinib
Relevant other oncological history	None	
Previous primary tumor	None	
Relevant non-oncological history	2023	Rheumatoid arthritis

Patient current details (20-Feb-2025)

Toxicities grade >= 2 or unknown	None
LVEF	50%
Known allergies	None
Recent surgeries	None

Tumor details (20-Feb-2025)

Measurable disease	Yes
Lesions	Liver, Lung
No lesions present	CNS, Brain, Bone, Lymph node

Active medication details

Medication	Administration route	Start date	Stop date	Dosage	Frequency
St. John's Wort	Oral	01-Feb-2023		300 MILLIGRAMS	1 / 2 DAYS

Blood transfusions

Product	Date
Erythrocyte	20-Sep-2024

Trial Matching Details

National trials that are open and potentially eligible (2 trials)

Trial	Cohort	Molecular	Sites
EGFR-C797S-TRIAL (Phase 2)	EGFR C797S	EGFR C797S	NL: Tilburg, Germany: Stuttgart
EGFR-L858R-TRIAL (Phase 1)	EGFR L858R	EGFR L858R	NL: Tilburg, Germany: Stuttgart

Trials in this table are matched solely on molecular event and tumor type (clinical data excluded).

International trials that are open and potentially eligible (2 trials)

Trial	Cohort	Molecular	Sites
EGFR-BE (Phase 1)	EGFR L858R	EGFR L858R	Belgium: Brussels
KRAS-G12C-TRIAL-DE (Phase 1)	KRAS G12C	KRAS G12C	Germany: Stuttgart

Trials in this table are matched solely on molecular event and tumor type (clinical data excluded).

Trials and cohorts that are potentially eligible, but are closed (1 trial)

Trial	Cohort	Molecular	Sites	Warnings
METC 01 IEMOEN (Phase 1)	Dose expansion - monotherapy	None		Has not exhausted SOC (at least platinum doublet remaining)

Trials and cohorts that are considered ineligible (4 cohorts from 3 trials)

Trial	Cohort	Molecular	Ineligibility reasons
METC 02 KAYRAS (Phase 1/2)	Dose expansion - monotherapy - Colorectum	KRAS G12D, PD-L1 >= 50.0	No colorectal cancer
METC 03 NO-SEE797ES	Dose escalation - monotherapy	EGFR C797S	C797S in EGFR in canonical transcript
METC 05 PICKME3CA	<i>Applies to all cohorts below</i>	None	No PIK3CA activating mutation(s)

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Trial	Cohort	Molecular	Ineligibility reasons
	Dose expansion - monotherapy - NSCLC (closed)		
	Dose expansion - monotherapy - Other cancer types (closed)		Tumor belongs to DOID term(s) lung non-small cell carcinoma

Trials and cohorts that are not evaluable or ignored (0 trials)

None

Potentially eligible open trials & cohorts

METC 01

Potentially eligible	Yes
Acronym	IEMOEN
Title	Phase I first-in-human study to evaluate safety of IEMOEN, a new PD-L1 inhibitor in advanced solid tumors

Reference	Evaluation
I-03	WARN Has not exhausted SOC (at least platinum doublet remaining)
E-02	UNDETERMINED No measurement found for hemoglobin
E-03	UNDETERMINED No measurement found for absolute neutrophil count
E-01	PASS Has no other condition belonging to category autoimmune disease
I-01	PASS Patient is at least 18 years old

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Reference	Evaluation
I-02	PASS Has solid primary tumor Stage IV is considered metastatic

METC 01 - Dose escalation - monotherapy

Cohort ID	A
Potentially eligible?	Yes
Open for inclusion?	Yes
Has slots available?	No

METC 01 - Dose expansion - monotherapy

Cohort ID	B
Potentially eligible?	Yes
Open for inclusion?	No
Has slots available?	No

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METC 02

Potentially eligible **Yes**
Acronym **KAYRAS**
Title **A phase 1/2 trial for first in-human usage of KAYRAS, a new specific KRAS G12D inhibitor in NSCLC and colorectal cancer**

Reference	Evaluation
I-04	WARN Variant(s) G12D in KRAS but subclonal likelihood of > 50%
I-03	UNDETERMINED ASAT and ALAT are not present or cannot be evaluated
I-01	PASS Patient is at least 18 years old
I-02	PASS Stage IV is considered metastatic
I-05	PASS PD-L1 expression above minimum of 50.0

METC 02 - Dose expansion - monotherapy - NSCLC

Cohort ID **A**
Potentially eligible? **Yes**
Open for inclusion? **Yes**
Has slots available? **Yes**

Reference	Evaluation
I-02	PASS

Tumor belongs to DOID term(s) lung non-small cell carcinoma

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METC 02 - Dose expansion - monotherapy - Colorectum

Cohort ID

B

Potentially eligible?

No

Open for inclusion?

Yes

Has slots available?

Yes

Reference	Evaluation
I-02	FAIL No colorectal cancer

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METC 04

Potentially eligible **Yes**
Acronym **TEDR1**
Title **TEDR1 Trial: A phase II trial to evaluate efficacy of specific EGFR inhibitors in lung cancer**

Reference	Evaluation
I-1	PASS Patient is at least 18 years old
I-2	PASS Stage IV is considered metastatic Tumor belongs to DOID term(s) lung cancer
I-3	PASS C797S in EGFR in canonical transcript

METC 04 - Lung cancer C797S cohort

Cohort ID **A**
Potentially eligible? **Yes**
Open for inclusion? **Yes**
Has slots available? **Yes**

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Other trials & cohorts

METC 03

Potentially eligible	No
Acronym	NO-SEE797ES
Title	Phase I trial for development of NO-SEE797ES, a specific inhibitor for EGFR with C797 mutations but not C797S in solid tumors

Reference	Evaluation
I-03	FAIL C797S in EGFR in canonical transcript

METC 03 - Dose escalation - monotherapy

Cohort ID	A
Potentially eligible?	No
Open for inclusion?	Yes
Has slots available?	Yes

METC 05

Potentially eligible	No
Acronym	PICKME3CA
Title	A phase 1/2 trial of ABC123 +/- platinum doublet in PIK3CA-mutated solid cancer

Reference	Evaluation
I-04	FAIL No PIK3CA activating mutation(s)

METC 05 - Dose expansion - monotherapy - NSCLC

Cohort ID	A
Potentially eligible?	No
Open for inclusion?	No
Has slots available?	Yes

METC 05 - Dose expansion - monotherapy - Other cancer types

Cohort ID	B
Potentially eligible?	No
Open for inclusion?	No
Has slots available?	Yes

Reference	Evaluation
I-03	FAIL Tumor belongs to DOID term(s) lung non-small cell carcinoma

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